

# Symplectic Topology

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Based on the book “Introduction to Symplectic Topology” by Dusa McDuff & Dietmar Salamon

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# I. FOUNDATIONS

## 1 Symplectic Linear Algebra

### 1.1 Symplectic spaces

#### Definition 1.1.1

Let  $V$  be a finite-dimensional vector space. The **kernel** of a 2-form  $\omega \in \Lambda^2(V^*)$  is the subspace

$$\ker \omega = \{v \in V \mid \omega(v, \bullet) = 0\}.$$

A 2-form is **nondegenerate** if its kernel is zero.

A nondegenerate 2-form is called a **symplectic form** and the pair  $(V, \omega)$  a **symplectic vector space**.

#### Example 1.1.2

The archetypal symplectic vector space is  $\mathbb{R}^{2n}$  equipped with the 2-form

$$\omega_0 = \sum_{i=1}^n dx_i \wedge dy_i$$

where  $x_1, \dots, x_n, y_1, \dots, y_n$  forms the standard basis and  $dx_i, dy_i$  form its dual basis.

It is clear that a 2-form  $\omega$  is nondegenerate iff its representation matrix relative to any basis is non-singular. The representation matrix relative to  $B = (v_1, \dots, v_n)$  is  $\omega_{ij} = \omega(v_i, v_j)$ . Equivalently, if the induced map  $\hat{\omega}: V \rightarrow V^*$  given by  $\hat{\omega}(v)(u) = \omega(v, u)$  is injective, equivalently an isomorphism.

#### Definition 1.1.3

Let  $(V, \omega_V)$  and  $(W, \omega_W)$  be two symplectic spaces. A **symplectomorphism** is a linear isomorphism  $\varphi: V \rightarrow W$  such that the pullback of  $\omega_W$  is  $\omega_V$ :  $\omega_V = \varphi^* \omega_W$ . Recall that the pullback is defined by

$$(\varphi^* \omega_W)(v, u) = \omega_W(\varphi v, \varphi u).$$

It is clear that the composition and inverse of symplectomorphisms are symplectomorphisms, so the category of symplectic spaces with symplectomorphisms forms a groupoid.

The collection of symplectomorphisms  $V \rightarrow V$  is a group denoted by  $\text{Sp}(V, \omega)$ .

#### Example 1.1.4

In the case of the standard symplectic structure on  $\mathbb{R}^{2n}$ , we use the notation  $\text{Sp}(2n) = \text{Sp}(\mathbb{R}^{2n}, \omega_0)$ . Notice that the representation of  $\omega_0$  relative to the standard basis is

$$J_0 = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}.$$

Thus  $\text{Sp}(2n)$  can be viewed as the subgroup of  $\text{GL}_{2n}(\mathbb{R})$  defined by

$$\text{Sp}(2n) = \{A \in \text{GL}_{2n}(\mathbb{R}) \mid A^\top J_0 A = J_0\}.$$

**Proposition 1.1.5**

$\mathrm{Sp}(2n)$  is a Lie group with Lie algebra

$$\mathfrak{sp}(2n) = \{-J_0 S \mid S \in \mathrm{Mat}_{2n} \text{ symmetric}\}.$$

**Proof**

Let  $\mathfrak{so}(2n) \leq \mathrm{Mat}_{2n}$  be the linear subspace of skew-symmetric matrices, and define the map

$$f: \mathrm{Mat}_{2n} \rightarrow \mathfrak{so}(2n), \quad f(A) = A^\top J_0 A.$$

Then  $\mathrm{Sp}(2n) = f^{-1}J_0$ , so to show that  $\mathrm{Sp}(2n)$  is a Lie subalgebra of  $\mathrm{GL}_{2n}(\mathbb{R})$ , it is sufficient to show that  $J_0$  is a regular value of  $f$ .

Note that  $f$ 's differential at  $A \in \mathrm{Mat}_{2n}$  is

$$df_A(X) = X^\top J_0 A + A^\top J_0 X.$$

If  $f(A) = J_0$  and  $B \in \mathfrak{so}(2n)$  (which is equal to  $\mathfrak{so}(2n)$ 's tangent space), then  $X = -\frac{1}{2}AJ_0B$  is a preimage of  $B$ . Thus  $df_A$  is surjective so that  $J_0$  is a regular value as desired.

Now,

$$\mathfrak{sp}(2n) = T_I \mathrm{Sp}(2n) = \ker df_I = \{X \in \mathrm{Mat}_{2n} \mid X^\top J_0 + J_0 X = 0\}.$$

Note that if  $X \in \mathfrak{sp}(2n)$  then  $X = -J_0 S$  where  $S = J_0 X$ , and this is symmetric:

$$S^\top = -X^\top J_0 = J_0 X = S.$$

Conversely for  $S$  symmetric,  $-J_0 S \in \mathfrak{sp}(2n)$ . ■

**Definition 1.1.6**

Let  $(V, \omega)$  be a symplectic space and  $W \leq V$  a subspace. Its **symplectic complement** is the subspace

$$W^\omega = \{v \in V \mid \omega(v, w) = 0 \text{ for all } w \in W\}.$$

Note that unlike the case for inner products,  $W^\omega$  and  $W$  need not be transversal. The subspace  $W$  is called

- (1) **isotropic** if  $W \leq W^\omega$ ,
- (2) **coisotropic** if  $W^\omega \leq W$ ,
- (3) **Lagrangian** if  $W = W^\omega$ ,
- (4) **symplectic** if  $W \cap W^\omega = 0$ .

Note that  $W$  is isotropic iff  $\omega$  vanishes on  $W$ , and symplectic iff  $(W, \omega|_W)$  is a symplectic space.

**Lemma 1.1.7**

For a subspace  $W \leq V$ ,

$$\dim W + \dim W^\omega = \dim V, \quad W^{\omega\omega} = W.$$

**Proof**

Consider  $\widehat{\omega}: V \rightarrow V^*$  which is an isomorphism by non-degeneracy. Let us define

$$\text{ann } W = \{\varphi \in V^* \mid \varphi(W) = 0\}.$$

Because  $\widehat{\omega}$  is an isomorphism, we have

$$\text{ann } W = \{\widehat{\omega}(v) \mid \widehat{\omega}(v)(W) = 0\} = \widehat{\omega}(W^\omega)$$

as desired.

Notice that a functional in  $\text{ann } W$  is the same as a functional over  $V/W$ . Indeed,  $\varphi \in \text{ann } W$  iff  $W \leq \ker \varphi$  iff we can quotient  $\varphi$  by  $W$ . Thus  $\text{ann } W \cong (V/W)^*$  which has dimension  $\dim V - \dim W$  as desired. ■

Let  $V$  be a  $2n$ -dimensional vector space with basis  $u_1, \dots, u_n, v_1, \dots, v_n$ . Then we can define a symplectic form on  $V$  by

$$\omega = \sum_{i=1}^n du_i \wedge dv_i,$$

equivalently

$$\omega(u_i, u_j) = \omega(v_i, v_j) = 0, \quad \omega(u_i, v_j) = \delta_{ij}, \quad [\omega] = J_0.$$

Conversely any basis of a symplectic space which represents the symplectic form like this is called a **symplectic basis**.

Notice that the existence of a symplectic basis is equivalent to the existence of a symplectomorphism  $(\mathbb{R}^{2n}, \omega_0) \rightarrow (V, \omega)$ .

**Theorem 1.1.8**

Let  $(V, \omega)$  be a symplectic space. Then  $V$  has even dimension,  $\dim V = 2n$ , and there exists a symplectic basis.

**Proof**

We induct on  $m = \dim V$ . For  $m = 0$  this is trivial, so suppose  $m > 0$ . Because  $\omega$  is non-degenerate there are  $u_1, v_1 \in V$  such that  $\omega(u_1, v_1) \neq 0$ . We can normalize this and assume  $\omega(u_1, v_1) = 1$ .

Because  $\omega$  is alternating, this means that  $u_1, v_1$  are linearly independent. Furthermore the subspace  $W$  spanned by  $u_1, v_1$  is symplectic as  $\omega$  is nondegenerate there. Thus  $V = W \oplus W^\omega$ , and by induction  $W^\omega$  has even dimension and a symplectic basis. By adding  $u_1, v_1$  to  $W^\omega$ 's symplectic basis we get a symplectic basis of  $V$ , and  $V$  has even dimension. ■

An immediate corollary of this theorem is the following.

**Corollary 1.1.9**

Two symplectic spaces  $V, W$  are symplectomorphic iff  $\dim V = \dim W$ .

**Corollary 1.1.10**

Let  $V$  be a  $2n$ -dimensional vector space and  $\omega$  a 2-form on  $V$ . Then  $\omega$  is nondegenerate iff its  $n$ -fold exterior power  $\omega^n$  is nonzero.

**Proof**

If  $\omega$  is degenerate then there is a  $v \neq 0$  such that  $v \lrcorner \omega = 0$ . Because interior multiplication is an antiderivation:

$$v \lrcorner \omega^n = \omega \wedge (v \lrcorner \omega^{n-1}) + (v \lrcorner \omega) \wedge \omega^{n-1} = \omega \wedge (v \lrcorner \omega^{n-1}).$$

Thus inductively we have  $v \lrcorner \omega^n = 0$ . Because  $\omega^n$  is a top-form this means  $\omega^n = 0$  (we can see this by extending  $v$  to a basis, then  $\omega$  on that basis is zero).

Conversely if  $\omega$  is nondegenerate, then by the previous theorem there is a symplectic basis

$$\omega = \sum_i dx_i \wedge dy_i.$$

Then

$$\omega^n = \sum_I dx_{i_1} \wedge dy_{i_1} \wedge \cdots \wedge dx_{i_n} \wedge dy_{i_n},$$

and we may remove  $I$ s with repeated indices, so the sum is over permutations of  $1, \dots, n$ . Each permutation gives the same product as 2-forms commute, thus

$$\omega^n = n! dx_1 \wedge dy_1 \wedge \cdots \wedge dx_n \wedge dy_n \neq 0. \quad \blacksquare$$

**Proposition 1.1.11**

The Lagrangian subspaces of  $V$  are precisely the maximal isotropic subspaces. Moreover, every basis of a Lagrangian subspace can be extended to a symplectic basis of  $V$ .

**Proof**

If  $W$  is isotropic then  $W \leq W^\omega$  meaning  $\dim W \leq \frac{1}{2} \dim V$ . Since Lagrangian subspaces have dimension equal to  $\frac{1}{2} \dim V$ , Lagrangian subspaces are maximal. Conversely if  $W$  is isotropic and  $W \neq W^\omega$ , then there is a  $v \in W^\omega - W$ . Adding  $v$  to  $W$  results in an isotropic subspace as  $\omega$  vanishes on  $W + v$ . Thus maximal isotropic subspaces must have  $W = W^\omega$  so that  $W$  is Lagrangian.

For the second point it is sufficient to consider  $V = \mathbb{R}^{2n}$  with the standard symplectic structure. Let  $\Lambda \leq V$  be Lagrangian, then  $\Lambda' = J_0 \Lambda$  is symplectic as well. Notice that  $\Lambda' \cap \Lambda = 0$  because for  $u \in \Lambda$  we have  $\omega_0(u, J_0 u) = -u^\top u \neq 0$ . Thus  $\mathbb{R}^{2n} = \Lambda' \oplus \Lambda$ .

Let  $u_1, \dots, u_n$  be a basis for  $\Lambda$ . Consider the map  $\hat{\omega}: \Lambda' \rightarrow \Lambda^*$ . This is injective as its kernel is  $\Lambda \cap \Lambda^\omega = 0$ , and thus defines an isomorphism. Let  $\varphi_1, \dots, \varphi_n$  be the dual basis of  $u_1, \dots, u_n$ , and thus there are  $v_1, \dots, v_n \in \Lambda'$  such that  $\varphi_i = \hat{\omega}(v_i)$ .

Because  $\Lambda, \Lambda'$  are Lagrangian we have that

$$\omega(u_i, u_j) = \omega(v_i, v_j) = 0, \quad \omega(v_i, u_j) = \varphi_i(u_j) = \delta_{ij}$$

as desired. \blacksquare

**Proposition 1.1.12**

Let  $W \leq V$  be coisotropic.

- (1) The quotient  $\overline{W} = W/W^\omega$  has a unique symplectic structure  $\overline{\omega}$  whose pullback under projection  $W \rightarrow \overline{W}$  agrees with  $\omega|_W$ .
- (2) If  $\Lambda \leq V$  is Lagrangian, then

$$\overline{\Lambda} = \frac{(\Lambda \cap W) + W^\omega}{W^\omega} \leq \overline{W}$$

is Lagrangian.

### Proof

Let  $[w] \in \overline{W}$  denote the coset of  $w$ . Notice that  $\overline{\omega}$  must satisfy

$$\omega(w, w') = \overline{\omega}([w], [w']),$$

which uniquely determines  $\overline{\omega}$ , showing uniqueness. Because  $W$  is coisotropic,  $W^\omega$  is an isotropic subspace of  $W$ , and so this is well-defined: for  $v, v' \in W^\omega$  and  $w, w' \in W$ ,

$$\omega(w + v, w' + v') = \omega(w, w') + \omega(v, w') + \omega(w, v') + \omega(v, v') = \omega(w, w').$$

The middle two terms are zero because  $v, v' \in W^\omega$ , and  $v \in W^\omega \leq W$  so the last term is zero. Furthermore  $\overline{\omega}$  is nondegenerate as if  $\overline{\omega}([w], \bullet) = 0$ , then  $w \in W^\omega$  so  $[w] = 0$ . So we have shown (1).

For (2) we show that

$$\tilde{\Lambda} = (\Lambda \cap W) + W^\omega \leq V$$

is Lagrangian. Indeed,

$$\tilde{\Lambda}^\omega = (\Lambda \cap W)^\omega \cap W = (\Lambda^\omega + W^\omega) \cap W = \Lambda \cap W + W^\omega = \tilde{\Lambda}.$$

And  $\tilde{\Lambda}$  quotients to  $\overline{\Lambda}$ , and  $\tilde{\Lambda}^\omega$  quotients to  $\overline{\Lambda}^\omega$ , which are then equal. ■

## 1.2 The symplectic group

Recall that we have defined a groupoid whose objects are symplectic spaces, and whose isomorphisms are the symplectomorphisms. And corollary 1.1.9 says that the dimension of a symplectic space is its only significant invariant. Let us denote by

$$\mathrm{Sp}(V, \omega) = \{\Psi \in \mathrm{GL}(V) \mid \Psi^* \omega = \omega\}$$

the automorphism group of  $(V, \omega)$ , called the **symplectic group**.

By proposition 1.1.5,  $\mathrm{Sp}(V, \omega)$  is a Lie group. Following this proposition, we have the following result.

### Theorem 1.2.1

The symplectic group  $\mathrm{Sp}(V, \omega)$  is a Lie group with Lie algebra

$$\mathfrak{sp}(V, \omega) = \{T \in \mathrm{end}(V) \mid \omega(A\bullet, \bullet) + \omega(\bullet, A\bullet) = 0\}.$$

We can understand this group by focusing on the case of  $(\mathbb{R}^{2n}, \omega_0)$ . Then  $\mathrm{Sp}(2n) = \mathrm{Sp}(\mathbb{R}^{2n}, \omega_0)$  may be thought of as a Lie group of matrices  $\Psi \in \mathrm{Mat}_{2n}$  satisfying  $\Psi^\top J_0 \Psi = J_0$ . Such matrices are called **symplectic**.

### Lemma 1.2.2

Symplectic matrices have determinant one.

### Proof

By corollary 1.1.10,  $\omega_0^n$  is nonzero and thus a volume form on  $\mathbb{R}^{2n}$ . Furthermore a symplectic matrix  $\Psi$  satisfies  $\Psi^* \omega_0 = \omega_0$  and thus  $\Psi^* \omega_0^n = \omega_0^n$ . Because this is a volume form,  $\det \Psi = 1$ . ■

## 6 The symplectic group

We identify  $\mathbb{R}^{2n}$  with  $\mathbb{C}^n$  by identifying  $(x, y) \in \mathbb{R}^{2n}$  with  $x + iy \in \mathbb{C}^n$ . Then  $J_0$  corresponds to multiplication by  $i$  in  $\mathbb{C}^n$ . With this identification,  $\mathrm{GL}_n(\mathbb{C})$  is the subgroup of  $\mathrm{GL}_{2n}(\mathbb{R})$  consisting of matrices which commute with  $J_0$ :

$$\mathrm{GL}_n(\mathbb{C}) = \{A \in \mathrm{GL}_{2n}(\mathbb{R}) \mid J_0 A = A J_0\}.$$

In this identification, a matrix  $X + iY \in \mathrm{Mat}_n(\mathbb{C})$  corresponds to the matrix

$$X + iY \in \mathrm{Mat}_n(\mathbb{C}) \leftrightarrow \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \in \mathrm{Mat}_{2n}(\mathbb{R}).$$

Notice how the conjugate transpose acts on  $\mathrm{Mat}_n(\mathbb{C})$ :

$$(X + iY)^\dagger = (X - iY)^\top = \begin{pmatrix} X^\top & Y^\top \\ -Y^\top & X^\top \end{pmatrix} = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}^\top.$$

Thus  $X + iY \in \mathrm{U}(n)$  iff

$$\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}^{-1} = \begin{pmatrix} X^\top & Y^\top \\ -Y^\top & X^\top \end{pmatrix},$$

which is iff

$$XX^\top + YY^\top = I, \quad XY^\top = YX^\top.$$

Notice that

$$i(X + iY)^\dagger(X + iY) = i \iff (X + iY)^\dagger = (X + iY)^{-1} \iff X + iY \in \mathrm{U}(n).$$

Thus we see

$$\mathrm{U}(n) = \mathrm{GL}_n(\mathbb{C}) \cap \mathrm{Sp}(2n).$$

In fact we can extend this result.

### Lemma 1.2.3

$$\mathrm{Sp}(2n) \cap \mathrm{O}(2n) = \mathrm{Sp}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{O}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{U}(n).$$

### Proof

A matrix  $\Psi \in \mathrm{GL}_{2n}(\mathbb{R})$  satisfies

$$\Psi \in \mathrm{GL}_n(\mathbb{C}) \iff \Psi J_0 = J_0 \Psi, \quad \Psi \in \mathrm{Sp}(2n) \iff \Psi^\top J_0 \Psi = J_0, \quad \Psi \in \mathrm{O}(2n) \iff \Psi^\top \Psi = I.$$

Note that any of these two conditions imply the third, so we have the first two equalities. By the above discussion, we also get the final.  $\blacksquare$

For a linear operator  $\Psi$  we denote its spectrum by  $\sigma(\Psi)$ ; this is its set of eigenvalues.

### Lemma 1.2.4

Let  $\Psi \in \mathrm{Sp}(2n)$ , then  $\lambda \in \sigma(\Psi)$  iff  $\lambda^{-1} \in \sigma(\Psi)$ , and the multiplicities of  $\lambda$  and  $\lambda^{-1}$  are equal. If  $\pm 1$  is an eigenvalue of  $\Psi$  then it has even multiplicity.

Finally, if  $\lambda, \lambda' \in \sigma(\Psi)$  are two (not necessarily distinct) eigenvalues such that  $\lambda\lambda' \neq 1$ , then  $\omega_0(z, z') = 0$  for any eigenvectors of eigenvalues  $\lambda, \lambda'$  respectively.

### Proof

The first statement is due to the fact that  $\Psi^{-1}$  is similar to  $\Psi^\top$ :

$$\Psi^\top = J_0 \Psi^{-1} J_0^{-1},$$

$\Psi$  and  $\Psi^\top$  share the same eigenvalues (with the same multiplicities), and  $\lambda$  is an eigenvalue of  $\Psi$  iff  $\lambda^{-1}$  is of  $\Psi^{-1}$  (and the multiplicities agree).

Thus the total multiplicity of the eigenvalues not equal to  $\pm 1$  is even (as they come in pairs). The determinant of  $\Psi$  is the product of eigenvalues, as they come in pairs with their inverses this is equal to  $(-1)^m$  where  $m$  is  $-1$ 's multiplicity. By lemma 1.2.2 this is equal to 1 so  $m$  is even. Because  $\Psi$  has dimension  $2n$ , the multiplicity of 1 must be even as well.

The last statement is because

$$\omega_0(z, z') = \omega_0(\Psi z, \Psi z') = \lambda \lambda' \omega_0(z, z')$$

and since  $\lambda \lambda' \neq 1$ ,  $\omega_0(z, z') = 0$ . ■

Suppose  $D$  is a diagonal matrix,  $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ , then define

$$D^\alpha = \text{diag}(\lambda_1^\alpha, \dots, \lambda_n^\alpha)$$

for any real  $\alpha \in \mathbb{R}$  where this is defined. Suppose  $A \in \text{Mat}_n(\mathbb{R})$  is a diagonalizable matrix with  $A = P^{-1}DP$ ,  $D$  diagonal. Then define

$$A^\alpha = P^{-1}D^\alpha P$$

for real  $\alpha$  where this is defined. Note that this agrees with  $A^n$  for  $n$  natural. It is easy to see that this is well-defined (i.e. not dependent on  $P, D$ ).

If  $A^\alpha$  exists and  $Av = \lambda v$ , then  $A^\alpha v = \lambda^\alpha v$ . This is clear for diagonal  $A$ , and extends to diagonalizable  $A$  by extending  $v$  to an eigenbasis and using that  $P$ .

Note that if  $A$ 's eigenvalues are all positive, then this is defined for all  $\alpha$ . In particular if  $S$  is symmetric and positive definite, then its powers exist for all real  $\alpha$ .

#### Lemma 1.2.5

Exponentiation is continuous as a map

$$\text{Sym}_+ \times \mathbb{R} \rightarrow \text{Sym}_+, \quad (S, \alpha) \mapsto S^\alpha$$

where  $\text{Sym}_+ \subseteq \text{Mat}_n$  is the subspace of positive-definite symmetric matrices.

#### Proof

Let  $S \in \text{Sym}_+$ , and let its minimum and maximum eigenvalues be  $\lambda_{\min}, \lambda_{\max}$  respectively. Then we know for all vectors  $v$ ,

$$\lambda_{\min} |v|^2 \leq v^\top S v \leq \lambda_{\max} |v|^2$$

and these bounds are tight. We claim that there is an open neighborhood  $S \in U \subseteq \text{Sym}_+$  and numbers  $0 < m < M$  such that every  $A \in U$  has eigenvalues bound by  $m, M$ .

Let  $m = \lambda_{\min}/2$ , then for every symmetric matrix  $A$  and  $v$  of unit norm:

$$v^\top A v = v^\top S v + v^\top (A - S) v \geq \lambda_{\min} - |v^\top (A - S) v| \geq \lambda_{\min} - |(A - S)v| \geq \lambda_{\min} - |A - S|$$

where  $|A - S|$  denotes the operator norm. Thus if  $|A - S| < m$  then the right-hand side is equal to  $\lambda_{\min}/2$  and thus  $v^\top A v \geq \lambda_{\min}/2$ ; so all of  $A$ 's eigenvalues are bound from below by  $m$ . So all matrices in the ball of radius  $m$  around  $S$  have eigenvalues bound from below by  $m$ .

We can bound this from above too: for unit norm  $v$

$$v^\top A v \leq |A| \leq |S| + |A - S|,$$

and so if we require  $|A - S| < 1$  then

$$v^\top A v \leq |S| + 1 = \lambda_{\max} + 1,$$

so matrices in the ball of radius 1 around  $S$  have eigenvalues bound from above by  $M = \lambda_{\max} + 1$ . So the desired open set is

$$U = \{A \in \text{Sym}_+ \mid |A - S| < 1, m\}.$$

Instead of considering exponentiating over arbitrary  $\alpha$ , we show it is continuous for  $a \leq \alpha \leq b$  for any arbitrary  $a, b$ ; this is clearly sufficient. Consider the map  $f: [m, M] \times [a, b] \rightarrow \mathbb{R}$  by  $f(x, t) = t^x$ , this is continuous as  $m > 0$ . By Stone-Weierstrass, because the codomain is compact, we can approximate  $f$  uniformly by a sequence of polynomials  $p_k \in \mathbb{R}[x, t]$ .

Now suppose  $A \in U$ , so  $A = P^{-1} \text{diag}(\lambda_1, \dots, \lambda_n)P$ , and because  $p_k$  are polynomial

$$p_k(x, A) = P^{-1} \text{diag}(p_k(x, \lambda_1), \dots, p_k(x, \lambda_n))P.$$

Thus

$$A^\alpha - p_k(\alpha, A) = P^{-1} \text{diag}(\dots, \lambda_i - p_k(\alpha, \lambda_i), \dots)P.$$

Because  $A \in U$  the  $\lambda_i$  fall in  $[m, M]$ , and so under the operator norm

$$|A^\alpha - p_k(\alpha, A)| = \max_i |\lambda_i - p_k(\alpha, \lambda_i)| \leq \max_{t \in [m, M]} |t^\alpha - p_k(\alpha, t)|.$$

And so we see that the map  $F: U \times [a, b] \rightarrow \text{Sym}_+$ ,  $F(A, \alpha) = A^\alpha$ , satisfies for all  $A, \alpha$

$$|F(A, \alpha) - p_k(\alpha, A)| \leq \max_{t \in [m, M], \alpha \in [a, b]} |t^\alpha - p_k(\alpha, t)|,$$

so that  $(\alpha, A) \mapsto p_k(\alpha, A)$  uniformly converges to  $F$ . Because  $p_k$  are all polynomial and thus continuous, so too is  $F$ . ■

### Lemma 1.2.6

If  $S \in \text{Sp}(2n)$  is symmetric and positive definite, then  $P^\alpha \in \text{Sp}(2n)$  for all  $\alpha \in \mathbb{R}$ .

### Proof

As  $S$  is symmetric and positive definite we can decompose  $\mathbb{R}^{2n}$  into eigenspaces

$$\mathbb{R}^{2n} = \bigoplus_{i=1}^k E_i, \quad E_i = \ker(\lambda_i I - S),$$

for  $\lambda_i > 0$  the eigenvalues. For  $z, w \in \mathbb{R}^{2n}$  let their projections into  $E_i$  be denoted  $z_i, w_i$  respectively. Then because  $S$  is symplectic

$$\omega_0(z_i, w_j) = \omega_0(Pz_i, Pw_j) = \lambda_i \lambda_j \omega_0(z_i, w_j).$$

So either  $\lambda_i \lambda_j = 1$  or  $\omega_0(z_i, w_j) = 0$ , and thus Then

$$\omega_0(P^\alpha z, P^\alpha w) = \sum_{i,j} (\lambda_i \lambda_j)^\alpha \omega_0(z_i, w_j) = \sum_{i,j} \omega_0(z_i, w_j) = \omega_0(z, w)$$

so that  $P^\alpha \in \text{Sp}(2n)$ . ■

### Proposition 1.2.7

- (1)  $U(n)$  is a maximal compact subgroup of  $\text{Sp}(2n)$ .

(2) The inclusion  $U(n) \rightarrow \mathrm{Sp}(2n)$  is a strong deformation retract. In particular  $\mathrm{Sp}(2n)$  is connected.

### Proof

We first prove (2). Define

$$H: [0, 1] \times \mathrm{Sp}(2n) \rightarrow \mathrm{Sp}(2n), \quad H(t, \Psi) = \Psi(\Psi^\top \Psi)^{-t/2}.$$

This is continuous by lemma 1.2.5, as  $\Psi^\top \Psi$  is positive definite for any non-singular  $\Psi$ . This is well-defined by the previous lemma,  $H(\Psi, t) \in \mathrm{Sp}(2n)$  for all  $t$ .

Now

$$H_0 = \mathrm{id}, \quad H_t|_{U(n)} = \mathrm{id}$$

where the latter is because  $\Psi^\top \Psi = I$  for unitary  $\Psi$ . And for  $\Psi \in \mathrm{Sp}(2n)$ ,  $H_1(\Psi)$  is orthogonal:

$$H_1(\Psi)^\top H_1(\Psi) = (\Psi^\top \Psi)^{-1/2} \Psi^\top \Psi (\Psi^\top \Psi)^{-1/2} = I.$$

Because  $\mathrm{Sp}(2n) \cap O(2n) = U(n)$ , we have  $H_1(\mathrm{Sp}(2n)) = U(n)$ . Thus  $H$  is indeed a strong deformation retract. And so  $\mathrm{Sp}(2n)$  and  $U(n)$  are homotopy equivalent, and  $U(n)$  is connected so therefore  $\mathrm{Sp}(2n)$  is as well.

We show (1) now. Suppose  $G$  is a compact subgroup of  $\mathrm{Sp}(2n)$  containing  $U(n)$ , but not equal to  $U(n)$ . Then let  $\Psi \in G - U(n)$ , and so  $H_1(\Psi) = \Psi(\Psi^\top \Psi)^{-1/2} \in U(n) \subseteq G$ . Thus the positive definite symmetric matrix  $P = \Psi^\top \Psi$  belongs to  $G - U(n)$ . Therefore it must have an eigenvalue greater than 1: its eigenvalues come in pairs  $\lambda, \lambda^{-1}$ , so otherwise  $P = I$ . But then the sequence  $P, P^2, P^3, \dots$  in  $G$  has no convergent subsequence, contradicting  $G$ 's compactness. ■

### Remark 1.2.8

Every symplectic matrix  $\Psi \in \mathrm{Sp}(2n)$  can be uniquely written as  $\Psi = UP$  where  $U \in U(n)$  and  $P$  is a symmetric, positive definite, symplectic matrix. Existence is by

$$U = \Psi(\Psi^\top \Psi)^{-1/2}, \quad P = (\Psi^\top \Psi)^{1/2}.$$

Uniqueness is because the only symmetric positive definite unitary matrix is  $I$  (because its eigenvalues come in pairs, and thus must all be 1). If  $U \in U(n)$  can be written as  $U = U'P'$ , then  $P = U(U')^{-1}$  is unitary as well, and thus  $P = I$  so  $U = U'$ . So this is true for unitary matrices, and thus if  $UP = U'P'$  we have  $U = U'P'P^{-1}$  so  $U = U'$  and  $P'P^{-1} = I$  meaning  $P = P'$ . This shows uniqueness.

This decomposition is called **symplectic polar decomposition**.

### Proposition 1.2.9

The complex determinant map  $\det_{\mathbb{C}}: U(n) \rightarrow S^1$  induces an isomorphism of fundamental groups  $\pi_1(U(n)) \cong \pi_1(S^1) \cong \mathbb{Z}$ .

### Proof

The determinant map is a fibration with typical fiber  $SU(n)$ . Indeed for every  $e^{i\theta} \in S^1$  we can consider the open subset  $W$  of  $S^1$  consisting of numbers whose arguments are between  $\theta - \varepsilon$  and  $\theta + \varepsilon$ . Then  $\det_{\mathbb{C}}^{-1}(W) \rightarrow SU(n) \times W$  given by  $\Psi \mapsto (e^{i\varphi/n} \Psi, e^{i\varphi})$  where  $e^{i\varphi} = \det \Psi$  is a homeomorphism with inverse  $(\Phi, e^{i\varphi}) \mapsto e^{i\varphi/n} \Phi$ . It also commutes with  $\det_{\mathbb{C}}$  because  $\det(e^{i\varphi/n} \Phi) = e^{i\varphi}$ .

Thus there is an exact sequence

$$\pi_1(\mathrm{SU}(n)) \rightarrow \pi_1(\mathrm{U}(n)) \rightarrow \pi_1(S^1) \rightarrow \pi_0(\mathrm{SU}(n)).$$

It is sufficient then to show that  $\mathrm{SU}(n)$  is simply connected, and we get the desired result.

It is trivial that  $\mathrm{SU}(n)$  is simply connected for  $n = 1$ , as  $\mathrm{SU}(1) = \{1\}$ . For  $n \geq 2$  the map  $\mathrm{SU}(n) \rightarrow S^{2n-1}$  which maps  $U \in \mathrm{SU}(n)$  to its first column is a fibration with typical fiber  $\mathrm{SU}(n-1)$ . And so we have an exact sequence

$$\pi_1(\mathrm{SU}(n-1)) \rightarrow \pi_1(\mathrm{SU}(n)) \rightarrow \pi_1(S^{2n-1}) \rightarrow \pi_0(\mathrm{SU}(n-1)).$$

By induction this means  $\pi_1(\mathrm{SU}(n)) \cong \pi_1(S^{2n-1}) = 1$  because  $2n-1 > 1$ . ■